CUSTOMER CHURN ANALYSIS

Final Portfolio Report | Excel • SQL • Power BI

≈ 1,000	≈ 31%	≈ 38%
Customer records	Overall churn rate	Month-to-month churn

Objective

Analyze customer churn patterns, identify segments with higher observed churn, and recommend practical actions the business can investigate to improve retention.

This report is based on the project findings shared during the analysis and a structural review of the uploaded Power BI report file. Rates are marked approximate where exact final values were not independently extractable from the report file.

1. Executive Summary

The dataset contains approximately 1,000 customer records, with a reported overall churn rate of about 31%. The analysis examined churn by contract type, tenure, payment method, internet service, and monthly charges. The reported patterns suggest that month-to-month customers and customers in their first year deserve particular attention, while payment method, service type, and price level warrant further investigation.

The analysis is descriptive: these group differences identify where to investigate, but do not prove why customers leave. The recommended next step is to combine these patterns with cancellation reasons, customer support history, service quality, and controlled retention experiments.

2. Business Questions

• What share of customers churned?
• Which contract and tenure groups have higher churn rates?
• How do payment method and internet service relate to churn?
• Do customers with higher monthly charges show different churn patterns?
• Which retention actions should the business test first?

3. Data and Methodology

The project used a customer churn dataset reported as approximately 1,000 records and 16 columns. During data preparation, missing values and duplicate customer IDs were checked, and a spelling inconsistency involving “fiber optic” was corrected. Excel was used for cleaning and exploratory analysis, MySQL for business-question queries, and Power BI for measures, KPI cards, charts, and slicers.

Methods included grouped churn-rate comparisons, pivot tables, SQL aggregation, DAX measures, and interactive filtering. The uploaded Power BI file contains two pages: **Churn_Dashboard** and **Deep_Dive**. Page 1 includes five KPI cards, four churn-rate charts, and four slicers. Page 2 includes a churn-status donut chart, average monthly charges by churn status, and churn rate by monthly-charges group.

4. Key Findings

Area	Reported result	Business interpretation
Overall churn	≈ 31%	Baseline churn level for the dataset.
Contract type	Month-to-month ≈ 38%; one-year ≈ 23%; two-year ≈ 20%	Month-to-month customers have the highest reported churn rate.
Payment method	Electronic check ≈ 36%; bank transfer ≈ 29%; credit card ≈ 29%; mailed check ≈ 25%	Electronic-check customers have the highest reported rate among the listed methods.
Internet service	Fiber optic ≈ 36%	Reported churn is above the overall rate for this service group.
Monthly charges	Above $100 ≈ 36%	Higher-charge customers show elevated churn in the reported grouping.
Tenure	0–12 months ≈ 37.2%	Early-tenure customers may need closer onboarding and support.

5. Dashboard Review

The uploaded PBIX structure shows a clear two-page organization, with five KPI cards and four slicers on the overview page, and three visuals on the deep-dive page. The core visual choices fit the project goals: contract, tenure, payment method, and internet-service comparisons, plus churn status and monthly-charge analysis.

Edits recommended before publishing

1. Add missing chart titles. The Internet Service chart on Page 1 and all three visuals on Page 2 appear to have blank title text in the report definition. Use descriptive titles such as “Churn Rate by Internet Service,” “Customer Churn Status,” “Average Monthly Charges by Churn Status,” and “Churn Rate by Monthly Charges Group.”

2. Give the top row more breathing room. The title and slicers share the top band on Page 1. Check the rendered view in Power BI and ensure the title remains prominent and slicers are easy to read. If crowded, move slicers into a dedicated row or a compact left-side panel.

3. Validate the average calculation. The deep-dive chart is configured with a field labelled “Average of monthly_charges.” Confirm in Power BI that the Y-axis aggregation is **Average**, not Sum, and that the title matches the aggregation.

4. Confirm consistent churn-rate measures. KPI cards use measures from DAX_Measure_Table, while the segment charts reference a “Churn Rate” measure on Cleaned_Data. This may be intentional, but verify both measures calculate churned customers divided by total customers under the current filter context and return consistent results.

5. Test slicers on both pages. Confirm the intended slicers filter the Deep_Dive page if that is the intended user experience. Use View → Sync slicers if necessary.

6. Business Recommendations

Early-tenure retention: review onboarding, setup completion, first-month support contacts, and early satisfaction; test proactive check-ins and measure the result.

Month-to-month customers: investigate cancellation reasons and value perception; test suitable loyalty benefits or longer-term options without assuming contract length alone causes churn.

Electronic-check customers: examine payment setup, failed payments, and customer preferences; improve payment guidance and offer convenient alternatives where appropriate.

Higher monthly charges: review plan fit, features used, service experience, and satisfaction before testing plan adjustments or value-focused communication.

Fiber-optic service: investigate installation experience, outages, speed complaints, and support history before selecting an intervention.

7. Limitations

• The reported percentages are approximate and should be checked against the final workbook and dashboard before public release.
• The dashboard file structure can reveal visual configuration, but not independently verify all displayed runtime values from a rendered report.
• Segment-level rates should be accompanied by customer counts to show the size of each group.
• Contract type, tenure, monthly charges, and service type may overlap; interpret each pattern carefully.
• No causal claims or quantified financial impact are established by this analysis alone.

8. Next Analysis Steps

1. Add customer counts alongside segment churn rates.
2. Compare each segment with the overall baseline in percentage points.
3. Explore combinations such as contract type × tenure and service type × charges.
4. Analyze cancellation reasons and support/service records if available.
5. Estimate revenue at risk using verified counts and clearly stated assumptions.
6. Measure retention experiments against a baseline or control group.

9. Conclusion

The project highlights month-to-month and early-tenure customers as priority segments for further investigation. Payment method, internet service, and monthly charges also show patterns worth exploring. The most credible business response is to validate the results, investigate possible explanations, and test targeted retention actions while measuring outcomes.

10. Portfolio Checklist

■ Cleaned dataset (if permitted to share)
■ Excel workbook with cleaning checks, pivot tables, and charts
■ SQL script with business questions and queries
■ Power BI PBIX file with verified measures, titles, and working slicers
■ Final report with exact validated metrics
■ README covering objective, tools, methods, findings, and limitations
■ Dashboard screenshots for GitHub or portfolio use

Before publishing, apply the dashboard title and measure checks in Section 5 and replace approximate metrics with exact verified values.